Chapter 11: Vectors

Chapter 11 · Original slide 1 · Vectors

Y1 Chapter 11
Vectors

Chapter 11 · Original slide 2 · Chapter Overview

1: Add/scale factors and show vectors are parallel.

If $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$, determine $|\mathbf{a}|$.

4: Solve both geometric problems.

A donut has position vector $4\mathbf{i} + 3\mathbf{j}$...

If a ship moves at a velocity of $(12i + 5j)km/h$, what is its speed?

2: Calculate magnitude and direction of a vector.

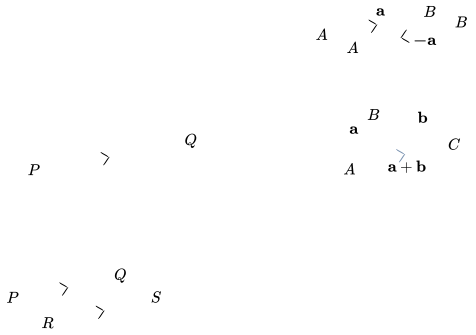
3: Understand and use position vectors.

An orienteer leaves position O and walks 15km at a bearing of 120° to the position A. Determine $\vec{OA}$.

5: Understand speed vs velocity

Chapter 11: Vectors

Chapter 11 · Original slide 3 · Vector Basics



A

Whereas a coordinate represents a position in space, a vector represents a displacement in space.

C

$\vec{AB} = -\vec{BA}$ and the two vectors are parallel, equal in magnitude but in opposite directions.

If P and Q are points then $\vec{PQ}$ is the vector between them.

B

If two vectors $\vec{PQ}$ and $\vec{RS}$ have the same magnitude and direction, they're the same vector and are parallel.

A vector has 2 properties:

- Direction
- Magnitude (i.e. length)

D

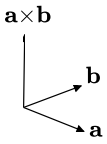
Triangle Law for vector addition:

$\vec{AB} + \vec{BC} = \vec{AC}$

The vector of multiple vectors is known as the resultant vector. (you will encounter this term in Mechanics)

Chapter 11 · Original slide 4 · Just for your interest…

Have you ever wondered what happens if you 'multiply' two vectors or two sets?



In KS2/3 you probably only experienced variables holding numerical values. You since saw that variables can represent other mathematical types, such as sets or vectors:

But when we do we need to define explicitly what operators like '$+$' and '$\times$' mean.

$$1 + 3 = 4$$

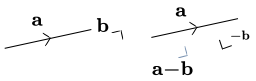
$$\begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

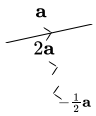
$$\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} =? \{a, b, c\} \times \{d, e\} =?$$

Often these operators are defined to give it properties that are consistent with its usage elsewhere, e.g. 'commutativity': $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ for vector addition just as $2 + 4 = 4 + 2$ for numbers.

Chapter 11: Vectors

Chapter 11 · Original slide 5 · Vector Basics





E

Vector subtraction is defined using vector addition and negation:

$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$

G

A scalar is a normal number, which can be used to 'scale' a vector.

• The direction will be the same.

• But the magnitude will be different (unless the scalar is 1).

If scalar is negative, opposite direction.

F

The zero vector $\mathbf{0}$ (a bold 0), represents no movement.

$\vec{PQ} + \vec{QP} = \mathbf{0}$

In 2D: $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

The implication is that if we can write one vector as a multiple of another, then we can show they are parallel.

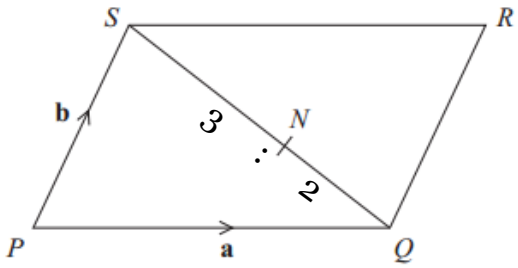
"Show $2\mathbf{a} + 4\mathbf{b}$ and $3\mathbf{a} + 6\mathbf{b}$ are parallel".

$3\mathbf{a} + 6\mathbf{b} = \frac{3}{2}(\mathbf{a} + 2\mathbf{b})$ ∴parallel

H

Any vector parallel to the vector $\mathbf{a}$ can be written as $\lambda\mathbf{a}$, where λ is a scalar.

Chapter 11 · Original slide 6 · Example



Tip: This ratio wasn't in the original diagram. I like to add the ratio as a visual aid.

a

For (b), there's two possible paths to get from N to R: via S or via Q. But which is best?

$PQRS$ is a parallelogram.

N is the point on SQ such that $SN : NQ = 3 : 2$.

$\overrightarrow{PQ} = \mathbf{a} \quad \overrightarrow{PS} = \mathbf{b}$

(a) Write down, in terms of $\mathbf{a}$ and $\mathbf{b}$, an expression for $\overrightarrow{SQ}$.

(b) Express $\overrightarrow{NR}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

b

$\vec{QR}$ is also b because it is exactly the same movement as $\vec{PS}$.

Tip: start your working with the first line so that your chosen route is clearer.

Chapter 11: Vectors

Chapter 11 · Original slide 7 · Test Your Understanding

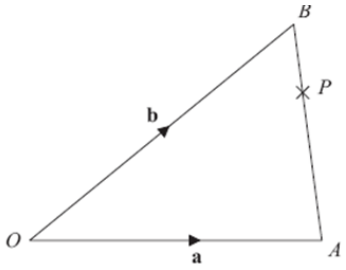
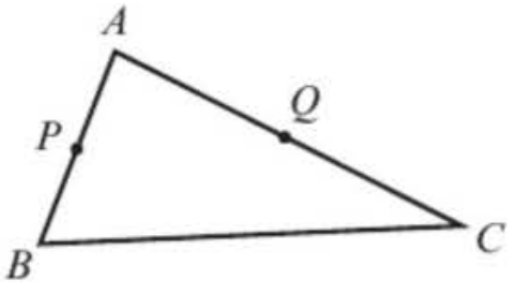


Diagram **NOT** accurately drawn

Chapter 11 · Original slide 8 · Exercise 11A - Year 1 Pages 234-235



9 In triangle ABC, $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AC} = \mathbf{b}$.

P is the midpoint of AB and Q is the midpoint of AC.

a Write in terms of **a** and **b**:

i $\overrightarrow{BC}$ **ii** $\overrightarrow{AP}$ **iii** $\overrightarrow{AQ}$ **iv** $\overrightarrow{PQ}$ **b** Show that PQ is parallel to BC.

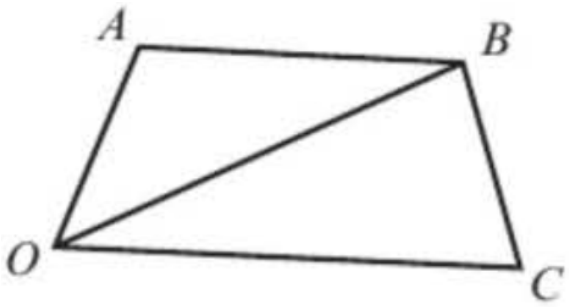
6 $ABCD$ is a trapezium with AB parallel to DC and $DC = 3AB$. M divides DC such that $DM : MC = 2 : 1$. $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{BC} = \mathbf{b}$. Find, in terms of **a** and **b**: **a** $\overrightarrow{AM}$

b $\overrightarrow{BD}$

c $\overrightarrow{MB}$

d $\overrightarrow{DA}$.

Chapter 11 · Original slide 9 · Exercise 11A - Year 1 Pages 234-235



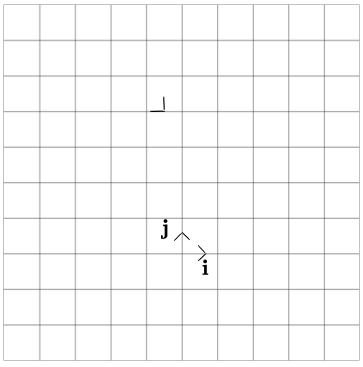
10 $OABC$ is a quadrilateral. $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = 3\mathbf{b}$ and $\overrightarrow{OB} = \mathbf{a} + 2\mathbf{b}$.

a Find, in terms of **a** and **b**:

i $\overrightarrow{AB}$ **ii** $\overrightarrow{CB}$ **b** Show that AB is parallel to OC.

11 The vectors $2\mathbf{a} + k\mathbf{b}$ and $5\mathbf{a} + 3\mathbf{b}$ are parallel. Find the value of k.

Chapter 11 · Original slide 10 · Representing Vectors



You should already be familiar that the value of a vector is the displacement in the x and y direction (if in 2D).

$\begin{pmatrix} 3 \\ -2 \end{pmatrix}$

$\mathbf{a} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$

$\mathbf{a} + \mathbf{b} =$

$2\mathbf{a} =$

! A unit vector is a vector of magnitude 1.

i and **j** are unit vectors in the x-axis and y-axis respectively.

$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Notes: This allows us to write any vector algebraically without using vector notation. Any point in 2D space, as a vector from the origin, can be obtained using a linear combination of **i** and **j**, e.g. if $P\,(5, -1)$, $\vec{OP} = 5\mathbf{i} - \mathbf{j}$. For this reason, **i** and **j** are known as basis vectors of 2D coordinate space. In fact, any two non-parallel/non-zero

Chapter 11 · Original slide 11 · Examples

If $\mathbf{a} = 3\mathbf{i}, \mathbf{b} = \mathbf{i} + \mathbf{j}, \mathbf{c} = \mathbf{i} - 2\mathbf{j}$ then:

1) Write $\mathbf{a}$ in vector form.

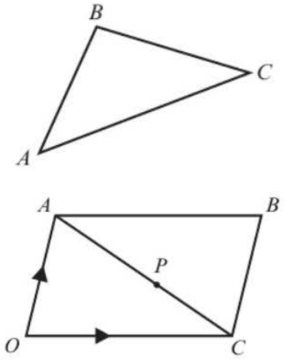
2) Find $\mathbf{b}+\mathbf{2c}$ in $\mathbf{i},\mathbf{j}$ form.

Chapter 11 · Original slide 12 · Exercise 11B - Year 1 Pages 237-238

4 Given that $\mathbf{a} = 2\mathbf{i} + 5\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - \mathbf{j}$, find:
a λ if $\mathbf{a} + \lambda\mathbf{b}$ is parallel to the vector $\mathbf{i}$.
b μ if $\mu\mathbf{a} + \mathbf{b}$ is parallel to the vector $\mathbf{j}$.

5 Given that $\mathbf{c} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{d} = \mathbf{i} - 2\mathbf{j}$, find:
a λ if $\mathbf{c} + \lambda\mathbf{d}$ is parallel to $\mathbf{i} + \mathbf{j}$.
b μ if $\mu\mathbf{c} + \mathbf{d}$ is parallel to $\mathbf{i} + 3\mathbf{j}$.
c s if $\mathbf{c} - s\mathbf{d}$ is parallel to $2\mathbf{i} + \mathbf{j}$.
d t if $\mathbf{d} - t\mathbf{c}$ is parallel to $-2\mathbf{i} + 3\mathbf{j}$.

Chapter 11 · Original slide 13 · Exercise 11B - Year 1 Pages 237-238



6 In triangle ABC, $\overrightarrow{AB} = 4\mathbf{i} + 3\mathbf{j}$ and $\overrightarrow{AC} = 5\mathbf{i} + 2\mathbf{j}$. Find $\overrightarrow{BC}$.

7 $OABC$ is a parallelogram. P divides AC in the ratio $3 : 2$. $\overrightarrow{OA} = 2\mathbf{i} + 4\mathbf{j}$, $\overrightarrow{OC} = 7\mathbf{i}$. Find in $\mathbf{i}, \mathbf{j}$ format and column vector format: **a** $\overrightarrow{AC}$
b $\overrightarrow{OP}$
c $\overrightarrow{AP}$.

Chapter 11 · Original slide 14 · Exercise 11B - Year 1 Pages 237-238

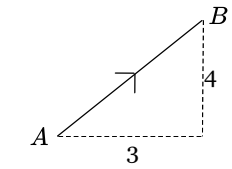
8 $\mathbf{a} = \begin{pmatrix} j \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 10 \\ k \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$. Given that $\mathbf{b} - 2\mathbf{a} = \mathbf{c}$, find the values of j and k.

9 $\mathbf{a} = \begin{pmatrix} p \\ -q \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} q \\ p \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$. Given that $\mathbf{a} + 2\mathbf{b} = \mathbf{c}$, find the values of p and q.

10 The resultant of the vectors $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{b} = p\mathbf{i} - 2p\mathbf{j}$ is parallel to the vector $\mathbf{c} = 2\mathbf{i} - 3\mathbf{j}$. Find:
a the value of p (4 m
b the resultant of vectors **a** and **b**. (1

Chapter 11: Vectors

Chapter 11 · Original slide 15 · Magnitude of a Vector





! The magnitude $|a|$ of a vector a is its length.

$\begin{pmatrix}3\\4\end{pmatrix}$

! If $\mathbf{a} = \begin{pmatrix}x\\y\end{pmatrix}$ $|\mathbf{a}| = \sqrt{x^2+y^2}$

$\mathbf{a} = \begin{pmatrix}4\\-1\end{pmatrix}$ $|\mathbf{a}| =$

$\mathbf{b} = \begin{pmatrix}2\\0\end{pmatrix}$ $|\mathbf{b}| =$

Chapter 11 · Original slide 16 · Unit Vectors

! A unit vector is a vector whose magnitude is 1

There's certain operations on vectors that require the vectors to be 'unit' vectors. We just scale the vector so that its magnitude is now 1.

$\mathbf{a} = \begin{pmatrix}3\\4\end{pmatrix}$

$|\mathbf{a}| =$

$\hat{\mathbf{a}} =$

If $\mathbf{a}$ is a vector, then the unit vector $\hat{\mathbf{a}}$ in the same direction is

$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$

Test Your Understanding: Convert the following vectors to unit vectors.

$\mathbf{a} = \begin{pmatrix}12\\-5\end{pmatrix}$

$|\mathbf{a}| =$

$\therefore \hat{\mathbf{a}} =$

$\mathbf{b} = \begin{pmatrix}1\\1\end{pmatrix}$

$|\mathbf{b}| =$

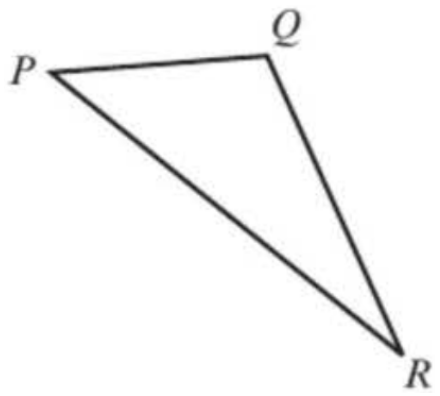
$\therefore \hat{\mathbf{b}} =$

Chapter 11 · Original slide 17 · Exercise 11C - Year 1 Pages 240-242

8 Given that $|2\mathbf{i} - k\mathbf{j}| = 2\sqrt{10}$, find the exact value of k.

9 Vector $\mathbf{a} = p\mathbf{i} + q\mathbf{j}$ has magnitude 10 and makes an angle θ with the positive x-axis where $\sin\theta = \frac{3}{5}$. Find the possible values of p and q.

Chapter 11 · Original slide 18 · Exercise 11C - Year 1 Pages 240-242



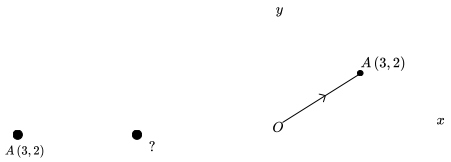
11 In triangle PQR, $\overrightarrow{PQ} = 4\mathbf{i} + \mathbf{j}$, $\overrightarrow{PR} = 6\mathbf{i} - 8\mathbf{j}$.

a Find the size of $\angle QPR$, in degrees, to one decimal place. (5 marks)

b Find the area of triangle PQR. (2 marks)

Chapter 11: Vectors

Chapter 11 · Original slide 19 · Position Vectors



Suppose we started at a point $(3, 2)$ and translated by the vector $\begin{pmatrix}4\\0\end{pmatrix}$:

$\mathbf{a} = \begin{pmatrix}3\\2\end{pmatrix}$

$\begin{pmatrix}4\\0\end{pmatrix}$

A vector used to represent a position is unsurprisingly known as a position vector.

A position can be thought of as a translation from the origin, as per above. It enables us to use positions in all sorts of vector (and matrix!) calculations.

You might think we can do something like:

$(3, 2) + \begin{pmatrix}4\\0\end{pmatrix} = (7, 2)$

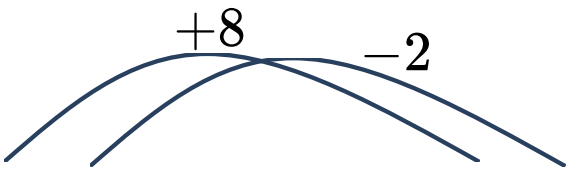
But only vectors can be added to other vectors.

If we treated the point $(\mathbf{3,2})$ as a vector, then this solves the problem:

$\begin{pmatrix}3\\2\end{pmatrix} + \begin{pmatrix}4\\0\end{pmatrix} = \begin{pmatrix}7\\2\end{pmatrix}$

! The position vector of a point A is the vector$\vec{OA}$, where O is the origin. $\vec{OA}$ is usually written as $\mathbf{a}$.

Chapter 11 · Original slide 20 · Example



The points A and B have coordinates $(3, 4)$ and $(11, 2)$ respectively.

Find, in terms of i and j:

a) The position vector of A

b) The position vector of B

c) The vector $\vec{AB}$

a b c

You can see this by inspection of the change in x and the change in y:

$A\,(3, 4) \rightarrow B(11, 2)$

More formally:

$\vec{AB} = \vec{OB} - \vec{OA}$

$= \begin{pmatrix}11\\2\end{pmatrix} - \begin{pmatrix}3\\4\end{pmatrix} = \begin{pmatrix}8\\-2\end{pmatrix}$

Chapter 11 · Original slide 21 · Further Example

a b

$\vec{OA} = 5i - 2j$ and $\vec{AB} = 3i + 4j$. Find:

a) The position vector of B.

b) The exact value of $\left|\vec{OB}\right|$ in simplified surd form.

Either a quick sketch will help you see this, or thinking of $\vec{OA}$ as the original position and $\vec{AB}$ as the translation.

Chapter 11 · Original slide 22 · Exercise 11D - Year 1 Pages 243-244

4 $OABCDE$ is a regular hexagon. The points A and B have position vectors **a** and **b** respectively, where O is the origin. Find, in terms of **a** and **b**, the position vectors of **a** C
b D
c E.

5 The position vectors of 3 vertices of a parallelogram

are $\begin{pmatrix}4\\2\end{pmatrix}$, $\begin{pmatrix}3\\5\end{pmatrix}$ and $\begin{pmatrix}8\\6\end{pmatrix}$.

Find the possible position vectors of the fourth vertex.

Chapter 11 · Original slide 23 · Exercise 11D - Year 1 Pages 243-244

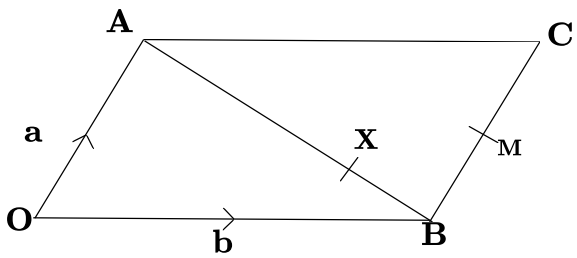
6 Given that the point A has position vector $4\mathbf{i} - 5\mathbf{j}$ and the point B has position vector $6\mathbf{i} + 3\mathbf{j}$,

a find the vector $\overrightarrow{AB}$. (2 ma

b find $|\overrightarrow{AB}|$ giving your answer as a simplified surd. (2 ma

7 The point A lies on the circle with equation $x^2 + y^2 = 9$. Given that $\overrightarrow{OA} = 2k\mathbf{i} + k\mathbf{j}$, find the exact value of k.

Chapter 11 · Original slide 24 · Solving Geometric Problems



X is a point on AB such that $AX : XB = 3 : 1$. M is the midpoint of BC.

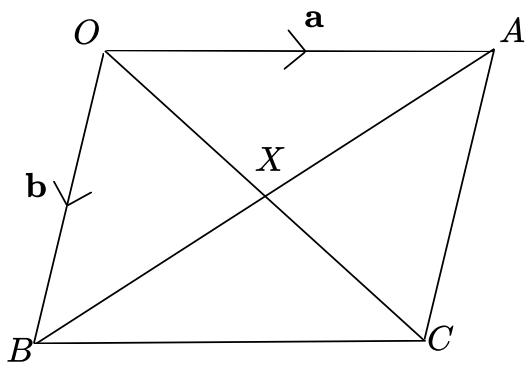
Show that $\vec{XM}$ is parallel to $\vec{OC}$.

For any proof question always find the vectors involved first, in this case $\vec{XM}$ and $\vec{OC}$.

The magic words here are "*is a multiple of*".

The key is to factor out a scalar such that we see the same vector.

Chapter 11 · Original slide 25 · Introducing Scalars and Comparing Coefficients



Remember when we had identities like:
$ax^2 + 3x \equiv 2x^2 + bx$

we could compare coefficients, so that $a = 2$ and $3 = b$.

We can do the same with (non-parallel) vectors!

$OACB$ is a parallelogram, where $\vec{OA} = a$ and $\vec{OB} = b$. The diagonals OC and AB intersect at a point X. Prove that the diagonals bisect each other.

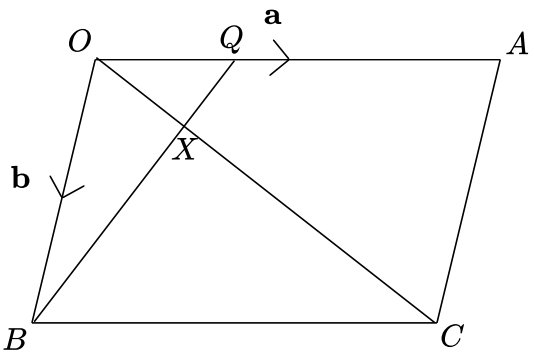
(Hint: Perhaps find $\vec{OX}$ in two different ways?)

We need to use a different scalar constant, this time μ.

It is common to use the letters λ and μ for scalars.

We don't know what fraction of the way across $\overrightarrow{BA}$ the point X is, so let λ be the fraction of the way across. We're hoping that $\lambda = \frac{1}{2}$, so that X is exactly halfway across and therefore bisects BA.

Chapter 11 · Original slide 26 · Test Your Understanding



In the above diagram, $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$ and $\vec{OQ} = \frac{1}{3}\mathbf{a}$. We wish to find the ratio $OX : XC$.

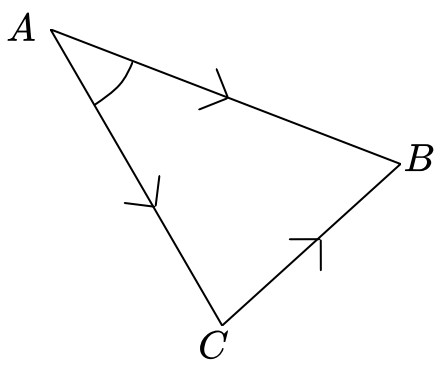
a) If $\vec{OX} = \lambda\vec{OC}$, find an expression for $\vec{OX}$ in terms of **a,b** and λ.

b) If $\vec{BX} = \mu\vec{BQ}$, find an expression for $\vec{OX}$ in terms of **a,b** and μ.

c) By comparing coefficients or otherwise, determine the value of λ, and hence the ratio $OX : XC$.

Expand and collect **a** terms and collect **b** terms, so that we can compare coefficients later.

Chapter 11 · Original slide 27 · Area of a Triangle



$\vec{AB} = 3\mathbf{i} - 2\mathbf{j}$ and $\vec{AC} = \mathbf{i} - 5\mathbf{j}$. Determine $\angle BAC$.

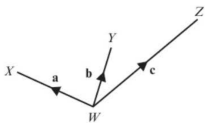
Strategy: Find 3 lengths of triangle then use cosine rule to find angle.

$\begin{pmatrix} 3 \\ -2 \end{pmatrix}$

$\begin{pmatrix} 1 \\ -5 \end{pmatrix}$

$\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

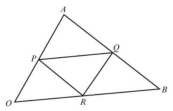
Chapter 11 · Original slide 28 · Exercise 11E - Year 1 Pages 246-247



1 In the diagram, $\overrightarrow{WX} = \mathbf{a}$, $\overrightarrow{WY} = \mathbf{b}$ and $\overrightarrow{WZ} = \mathbf{c}$. It is given that $\overrightarrow{XY} = \overrightarrow{YZ}$.

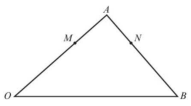
Prove that $\mathbf{a} + \mathbf{c} = 2\mathbf{b}$.

Chapter 11 · Original slide 29 · Exercise 11E - Year 1 Pages 246-247



2 OAB is a triangle. P, Q and R are the midpoints of OA, AB and OB respectively. OP and OR are equal to $\mathbf{p}$ and $\mathbf{r}$ respectively.
a Find **i** $\overrightarrow{OB}$ **ii** $\overrightarrow{PQ}$.
b Hence prove that triangle PAQ is similar to triangle OAB.

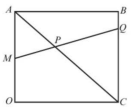
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3 OAB is a triangle. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. The point M divides OA in the ratio $2 : 1$. MN is parallel to OB.

a Express the vector $\overrightarrow{ON}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
b Show that $AN : NB = 1 : 2$.

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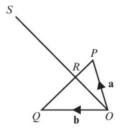


4 $OABC$ is a square. M is the midpoint of OA, and Q divides BC in the ratio $1 : 3$. AC and MQ meet at P.

a If $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$, express $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.
b Show that P divides AC in the ratio $2 : 3$.

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^{P}Q is a triangle.
$\overrightarrow{R} = \overrightarrow{RQ}$ and $3\overrightarrow{OR} = \overrightarrow{OS}$
$\vec{P} = \mathbf{a}$ and $\overrightarrow{OQ} = \mathbf{b}$.
Show that $\overrightarrow{OS} = 2\mathbf{a} + \mathbf{b}$.
Point T is added to the diagram such that $\overrightarrow{OT} = -\mathbf{b}$.
Prove that points T, P and S lie on a straight line.



6 OPQ is a triangle.

$2\overrightarrow{PR} = \overrightarrow{RQ}$ and $3\overrightarrow{OR} = \overrightarrow{OS}$ $\overrightarrow{OP} = \mathbf{a}$ and $\overrightarrow{OQ} = \mathbf{b}$.

a Show that $\overrightarrow{OS} = 2\mathbf{a} + \mathbf{b}$.

b Point T is added to the diagram such that $\overrightarrow{OT} = -\mathbf{b}$.

Chapter 11: Vectors

Chapter 11 · Original slide 33 · EXTENSION

1

[STEP 2010 Q7]

Relative to a fixed origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$, respectively. (The points O, A and B are not collinear.) The point C has position vector $\mathbf{c}$ given by

$\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$,

where α and β are positive constants with $\alpha + \beta < 1$. The lines OA and BC meet at the point P with position vector $\mathbf{p}$ and the lines OB and AC meet at the point Q with position vector $\mathbf{q}$. Show that

$\mathbf{p} = \frac{\alpha a}{1-\beta}$

and write down q in terms of α, β and b.

Show further that the point R with position vector r given by

$\mathbf{r} = \frac{\alpha\mathbf{a}+\beta\mathbf{b}}{\alpha+\beta}$,

lies on the lines OC and AB.

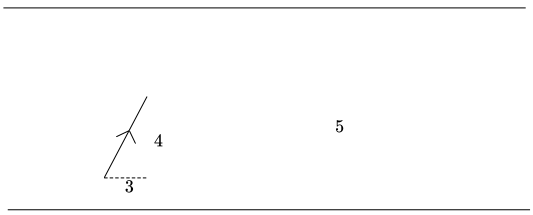
The lines OB and PR intersect at the point S. Prove that $\frac{OQ}{BQ} = \frac{OS}{BS}$.

Click here for the solution:

http://www.mathshelper.co.uk/STEP%202010%20Solutions.pdf

(go to Q7)

Chapter 11 · Original slide 34 · Modelling



In Mechanics, you will see certain things can be represented as a simple number (without direction), or as a vector (with direction):

Remember a ‘scalar’ just means a normal number (in the context of vectors). It can be obtained using the magnitude of the vector.

Vector Quantity

Equivalent Scalar Quantity

Velocity

e.g. $\begin{pmatrix}3\\4\end{pmatrix} km/h$

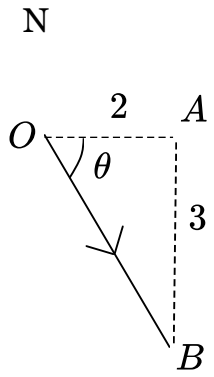
Speed

$= 5km/h$

This means the position vector of the object changes by $\begin{pmatrix}3\\4\end{pmatrix}$ each hour.

...which is equivalent to moving 5km each hour.

Chapter 11 · Original slide 35 · Example



[Textbook] A girl walks 2 km due east from a fixed point O to A, and then 3 km due south from A to B. Find

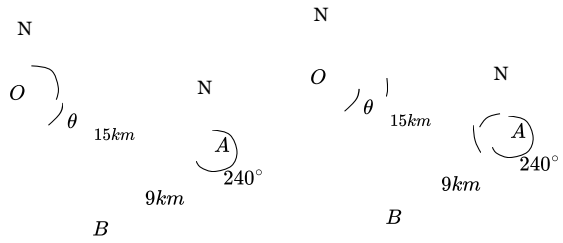
a) the total distance travelled

b) the position vector of B relative to O

c) $\left|\vec{OB}\right|$

d) The bearing of B from O.

Chapter 11 · Original slide 36 · Further Example



[Textbook] In an orienteering exercise, a cadet leaves the starting point O and walks 15 km on a bearing of 120° to reach A, the first checkpoint. From A he walks 9 km on a bearing of 240° to the second checkpoint, at B. From B he returns directly to O.

Find:

a) the position vector of A relative to O

b) $\left|\vec{OB}\right|$

c) the bearing of B from O

d) the position vector of B relative O.

b c

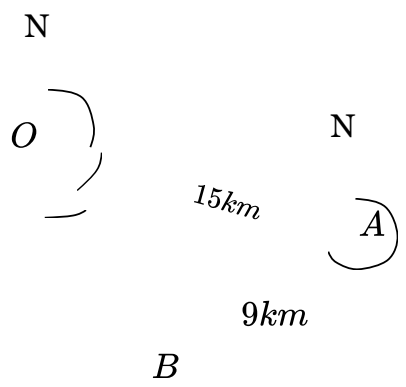
I have no specific advice to offer except:

1. Draw a BIG diagram.

2. Remember bearings are measured clockwise from North.

3. Don’t forget units (on vectors!)

Chapter 11 · Original slide 37 · Further Example



[Textbook] In an orienteering exercise, a cadet leaves the starting point O and walks 15 km on a bearing of 120° to reach A, the first checkpoint. From A he walks 9 km on a bearing of 240° to the second checkpoint, at B. From B he returns directly to O.

Find:

a) the position vector of A relative to O

b) $\left|\vec{OB}\right|$

c) the bearing of B from O

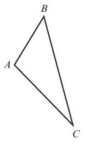
d) the position vector of B relative O.

Chapter 11 · Original slide 38 · Exercise 11F - Year 1 Pages 250-251

5 A particle P of mass $m = 0.3$ kg moves under the action of a single constant force $\mathbf{F}$ newtons. The acceleration of P is $\mathbf{a} = (5\mathbf{i} + 7\mathbf{j})$ m s^{-2}.
a Find the angle between the acceleration and $\mathbf{i}$. (2 ma
Force, mass and acceleration are related by the formula $\mathbf{F} = m\mathbf{a}$.
b Find the magnitude of $\mathbf{F}$. (3 ma

6 Two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$, are given by the vectors $\mathbf{F}_1 = (3\mathbf{i} - 4\mathbf{j})$ N and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})$ N. The resultant force, $\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2$ acts in a direction which is parallel to the vector $(2\mathbf{i} - \mathbf{j})$.
a Find the angle between $\mathbf{R}$ and the vector $\mathbf{i}$.
b Show that $p + 2q = 5$.
c Given that $p = 1$, find the magnitude of $\mathbf{R}$.

Chapter 11 · Original slide 39 · Exercise 11F - Year 1 Pages 250-251



7 The diagram shows a sketch of a field in the shape of a triangle ABC. Given $\overrightarrow{AB} = 30\mathbf{i} + 40\mathbf{j}$ metres and $\overrightarrow{AC} = 40\mathbf{i} - 60\mathbf{j}$ metres,

a find $\overrightarrow{BC}$ (2 ma
b find the size of $\angle BAC$, in degrees, to one decimal place (4 ma
c find the area of the field in square metres. (3 ma

Chapter 11 · Original slide 40 · Exercise 11F - Year 1 Pages 250-251

8 A boat has a position vector of $(2\mathbf{i} + \mathbf{j})$ km and a buoy has a position vector of $(6\mathbf{i} - 4\mathbf{j})$ km, relative to a fixed origin O.
a Find the distance of the boat from the buoy.
b Find the bearing of the boat from the buoy.
The boat travels with constant velocity $(8\mathbf{i} - 10\mathbf{j})$ km/h.
c Verify that the boat is travelling directly towards the buoy.
d Find the speed of the boat.
e Work out how long it will take the boat to reach the buoy.
Problem-solving Draw a sketch showing t initial positions of the bo the buoy and the origin.